

1. AGENT LAYER

Reasoning, MCP & Skills

2. INFRASTRUCTURE LAYER

Hybrid Execution & Supercompute

3. RESULT LAYER

Living Record & Provenance

Autonomous Agents

Protocols & Skills

Orbit

Scientific research workbench

Claude Code

Command-line agentic assistant

Codex

Autonomous code & task engine

Antigravity

Multimodal pair-programmer

Pi.dev

Headless agent runtime

...

Open to any MCP client

Galaxy MCP

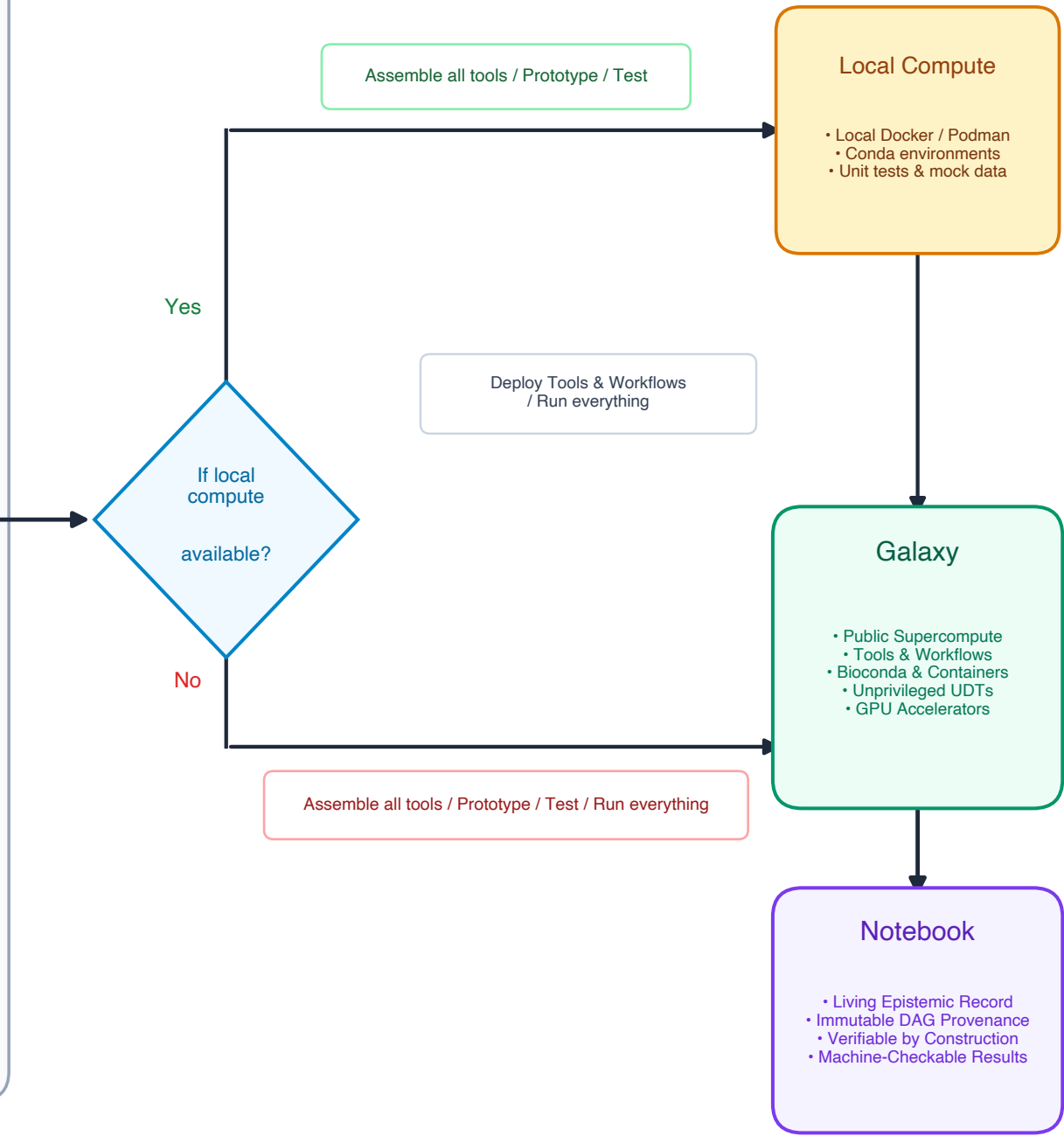
44 tool endpoints, stdio & HTTP

Galaxy Skills

Curated domain guides & idioms

Foundry Skills

Workflow casting & schema checks



Key Architectural Properties

Decoupled Execution

Agents reason and plan independently; computational workloads execute remotely across distributed clusters.

Universal Public Access

Researchers lacking local supercompute or GPU resources run heavy transformer models and structural aligners on Galaxy.

Dual-Mode Flexibility

Rapid local prototyping and unit-testing when workstations allow; direct remote assembly when local resources are absent.

Extensibility via UDTs

Unprivileged User-Defined Tools allow agents to cast custom containerized software dynamically at runtime.

Provenance by Construction

Every executed job automatically seals input digests, exact tool versions, and outputs into an immutable DAG ledger.